EXPERIMENT 10

PERFORMANCE ANALYSIS OF FM SYSTEM UNDER GAUSSIAN WHITE NOISE USING MATLAB

OBJECTIVE:

- To generate a Frequency Modulated (FM) signal in MATLAB using the specified carrier and message signal parameters.
- To introduce Additive White Gaussian Noise (AWGN) into the generated FM signal and simulate the effect of a noisy communication channel using MATLAB.
- To analyze the effect of AWGN on the transmitted FM signal by comparing the original and noisy FM signals using time-domain and frequency-domain analysis.

PROGRAM:

```
clc;

clear;

close all;


% Parameters

fs = 100000;

t = 0:1/fs:0.02;

fm = 500;       % Message frequency

fc = 5000;      % Carrier frequency

kf = 100;       % Frequency sensitivity


% Message Signal

m = cos(2*pi*fm*t);

c = cos(2*pi*fc*t);


% FM Modulation

fm_sig = cos(2*pi*fc*t + 2*pi*kf*cumsum(m)/fs);
```

```
% Add Gaussian White Noise (10 dB SNR)
snr_db = 10;
signal_power = mean(fm_sig.^2);
noise_power = signal_power / (10^(snr_db/10));
noise = sqrt(noise_power)*randn(size(fm_sig));
noisy = fm_sig + noise;

% Display
figure;

subplot(4,1,1)
plot(t,m)
title('Message Signal')
grid on

subplot(4,1,2)
plot(t,c)
title('Carrier Signal')
grid on

subplot(4,1,3)
plot(t,fm_sig)
title('FM Signal')
grid on
subplot(4,1,4)
plot(t,noisy)
title('FM Signal with Gaussian White Noise')
```

grid on

OUTPUT:

